Sheet 6 Solution - (PCA)

===

Question 1

Dataset (4 samples, 2 features)

Let the two features be $x_1$and x_2.

$$\boldsymbol{X} = \begin{bmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{3} & \mathbf{2} \\ \mathbf{4} & \mathbf{4} \\ \mathbf{5} & \mathbf{5} \end{bmatrix}$$

We will do PCA using **mean-cantering** (no standardization) because the two features are on a similar scale.

Step 1) Compute the mean of each feature

$$\mu = [\mu_{x_1} \quad \mu_{x_2}] = \left[\frac{2+3+4+5}{4} \quad \frac{1+2+4+5}{4}\right] = [3.5 \quad 3]$$

Step 2) Center the data

$$X_c = X - \mu$$

Row-by-row:

- For (2، 1): $(2-3.5, 1-3) = (-1.5, -2)$
- For (3، 2): $(-0.5, -1)$
- For (4، 4): $(0.5، 1)$
- For (5، 5): $(1.5، 2)$

So:

$$X_c = \begin{bmatrix} -1.5 & -2 \\ -0.5 & -1 \\ 0.5 & 1 \\ 1.5 & 2 \end{bmatrix}$$

Step 3) Compute the covariance matrix

For $n = 4$samples:

$$C = \frac{1}{n-1} X_c^T X_c = \frac{1}{3} X_c^T X_c$$

3.1 Variance of x_1

$$\mathrm{Var}(x_1) = \frac{1}{3}\sum(x_{1i} - \mu_{x_1})^2 = \frac{1}{3}((-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2)$$
$$= \frac{1}{3}(2.25 + 0.25 + 0.25 + 2.25) = \frac{5}{3} = 1.6667$$

3.2 Variance of x_2

$$\mathrm{Var}(x_2) = \frac{1}{3}((-2)^2 + (-1)^2 + (1)^2 + (2)^2) = \frac{1}{3}(4+1+1+4) = \frac{10}{3} = 3.3333$$

3.3 Covariance Cov(x_1, x_2)

$$\begin{aligned}\text{Cov}(x_1, x_2) &= \frac{1}{3}\sum(x_{1i} - \mu_{x_1})(x_{2i} - \mu_{x_2}) \\ &= \frac{1}{3}((-1.5)(-2) + (-0.5)(-1) + (0.5)(1) + (1.5)(2)) \\ &= \frac{1}{3}(3 + 0.5 + 0.5 + 3) = \frac{7}{3} = 2.3333\end{aligned}$$

So the covariance matrix is:

$$C = \begin{bmatrix} \frac{5}{3} & \frac{7}{3} \\ \frac{7}{3} & \frac{10}{3} \end{bmatrix} \approx \begin{bmatrix} 1.6667 & 2.3333 \\ 2.3333 & 3.3333 \end{bmatrix}$$

Step 4) Find eigenvalues (variance along principal directions)
Solve:

$$\det(C - \lambda I) = 0$$

$$\det\left(\begin{bmatrix} \frac{5}{3} - \lambda & \frac{7}{3} \\ \frac{7}{3} & \frac{10}{3} - \lambda \end{bmatrix}\right) = 0$$

$$\left(\frac{5}{3} - \lambda\right)\left(\frac{10}{3} - \lambda\right) - \left(\frac{7}{3}\right)^2 = 0$$

This simplifies to:

$$9\lambda^2 - 45\lambda + 1 = 0$$

So:

$$\lambda_{1,2} = \frac{15 \pm \sqrt{221}}{6}$$

Numerically:

- $\lambda_1 \approx 4.977678$(largest → **PC1**)
- $\lambda_2 \approx 0.022322$(small → **PC2**)

Explained variance ratio
Total variance $= \lambda_1 + \lambda_2 \approx 5$

$$\text{EVR}_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2} \approx \frac{4.977678}{5} = 0.995536 \ (\approx 99.55\%)$$
$$\text{EVR}_2 \approx 0.004464 \ (\approx 0.45\%)$$

Meaning: **almost all information is captured by PC1**.

Step 5) Find eigenvectors (principal directions)

PC1 eigenvector v_1
Solve $(C - \lambda_1 I)v_1 = 0$. One (unnormalized) solution is:

$$v_1 \propto \begin{bmatrix} \frac{14}{\sqrt{221}+5} \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0.7047 \\ 1 \end{bmatrix}$$

Normalize to unit length:

$$v_1 = \begin{bmatrix} 0.5760 \\ 0.8174 \end{bmatrix}$$

(Any sign flip [$-v_1$]is equivalent.)

PC2 eigenvector v_2(orthogonal to v_1)

$$v_2 = \begin{bmatrix} -0.8174 \\ 0.5760 \end{bmatrix}$$

Step 6) Project data onto the principal components (scores)

For each centered sample $x_{c,i}$, the PC1 score is:

$$z_{1,i} = x_{c,i} \cdot v_1$$

Compute each:

1. $x_c = (-1.5, -2)$
   $$z_1 = (-1.5)(0.5760) + (-2)(0.8174) = -0.8640 - 1.6348 = -2.4988 \approx -2.4989$$
2. $x_c = (-0.5, -1)$
   $$z_1 = (-0.5)(0.5760) + (-1)(0.8174) = -0.2880 - 0.8174 = -1.1054$$
3. $x_c = (0.5, 1)$
   $$z_1 = (0.5)(0.5760) + (1)(0.8174) = 0.2880 + 0.8174 = 1.1054$$
4. $x_c = (1.5, 2)$
   $$z_1 = (1.5)(0.5760) + (2)(0.8174) = 0.8640 + 1.6348 = 2.4988 \approx 2.4989$$

So, the **1D representation** (keeping only PC1) is:

$$z_1 = \begin{bmatrix} -2.4989 \\ -1.1054 \\ 1.1054 \\ 2.4989 \end{bmatrix}$$

(If you also project onto PC2: $z_{2,i} = x_{c,i} \cdot v_2$, you'll get very small values because $\lambda_2$is tiny.)

Step 7) (Optional but great for teaching) Reconstruct using only PC1

Approximate the original data using just PC1:

$$\hat{x}_i = \mu + z_{1,i} v_1^T$$

Example for the first point $z_1 = -2.4989$:

$$\hat{x} = [3.5 \quad 3] + (-2.4989)[0.5760 \quad 0.8174] = [3.5 - 1.4395 \quad 3 - 2.0426] = [2.0605 \quad 0.9574]$$

Reconstructed points (PC1 only) are approximately:

$$\hat{X} \approx \begin{bmatrix} 2.0605 & 0.9574 \\ 2.8632 & 2.0964 \\ 4.1368 & 3.9036 \\ 4.9395 & 5.0426 \end{bmatrix}$$

These are very close to the original X, showing **why PCA is good for compression/denoising**.

What students should conclude

- **PC1 explains ~99.55%** of variance → reducing from 2D to 1D loses very little information.
- The principal direction is:

$$v_1 = \begin{bmatrix} 0.5760 \\ 0.8174 \end{bmatrix}$$

- The PCA "new coordinate" (feature) for each sample is its score $z_{1,i}$.

Question 2

PCA Numerical Question 2 (4 samples, 2 features)
Given data

$$\boldsymbol{X} = \begin{bmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{1} & \mathbf{2} \\ \mathbf{-1} & \mathbf{-2} \\ \mathbf{-2} & \mathbf{-1} \end{bmatrix}$$

Step 1) Compute the mean

$$\mu = \left[\frac{2+1-1-2}{4}, \ \frac{1+2-2-1}{4}\right] = [0, \ 0]$$

So the data is already centered:

$$X_c = X$$

Step 2) Compute covariance matrix
Use:

$$C = \frac{1}{n-1} X_c^T X_c, n = 4 \Rightarrow C = \frac{1}{3} X^T X$$

First compute sums:
Variances

$$\sum x_1^2 = 2^2 + 1^2 + (-1)^2 + (-2)^2 = 4 + 1 + 1 + 4 = 10$$
$$\sum x_2^2 = 1^2 + 2^2 + (-2)^2 + (-1)^2 = 1 + 4 + 4 + 1 = 10$$

Covariance term

$$\sum x_1 x_2 = (2)(1) + (1)(2) + (-1)(-2) + (-2)(-1) = 2 + 2 + 2 + 2 = 8$$

So:

$$C = \frac{1}{3}\begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} & \frac{8}{3} \\ \frac{8}{3} & \frac{10}{3} \end{bmatrix}$$

Step 3) Find eigenvalues

For matrices of the form $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$:

- eigenvalues are $a + b$and $a - b$

Here $a = \frac{10}{3}$, $b = \frac{8}{3}$

$$\lambda_1 = a + b = \frac{18}{3} = 6$$

$$\lambda_2 = a - b = \frac{2}{3} = 0.6667$$

Explained variance ratio

Total variance:

$$\lambda_1 + \lambda_2 = 6 + \frac{2}{3} = \frac{20}{3}$$

$$\mathrm{EVR}_1 = \frac{6}{20/3} = \frac{18}{20} = 0.90(90\%)$$

$$\mathrm{EVR}_2 = \frac{2/3}{20/3} = \frac{2}{20} = 0.10(10\%)$$

PC1 captures **90%** of the variance.

Step 4) Find eigenvectors (principal directions)

For $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$:

- eigenvector for $a + b$is $[1, 1]^T$
- eigenvector for $a - b$is $[1, -1]^T$

Normalize them:

$$v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 5) Project data to get PCA scores

Scores:

$$\boldsymbol{z_{1,i} = x_i \cdot v_1, \qquad z_{2,i} = x_i \cdot v_2}$$

Because $\boldsymbol{v_1} = \frac{1}{\sqrt{2}}[\mathbf{1}, \mathbf{1}]$:

$$z_1 = \frac{x_1 + x_2}{\sqrt{2}}$$

and because $\boldsymbol{v_2} = \frac{1}{\sqrt{2}}[\mathbf{1}, -\mathbf{1}]$:

$$z_2 = \frac{x_1 - x_2}{\sqrt{2}}$$

Now compute for each point:

1) $\boldsymbol{x} = (\mathbf{2}, \mathbf{1})$

$$z_1 = \frac{2+1}{\sqrt{2}} = \frac{3}{\sqrt{2}} = 2.1213, z_2 = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}} = 0.7071$$

2) $\boldsymbol{x} = (\mathbf{1}, \mathbf{2})$

$$z_1 = \frac{3}{\sqrt{2}} = 2.1213, z_2 = \frac{-1}{\sqrt{2}} = -0.7071$$

3) $\boldsymbol{x} = (-\mathbf{1}, -\mathbf{2})$

$$z_1 = \frac{-3}{\sqrt{2}} = -2.1213, z_2 = \frac{(-1)-(-2)}{\sqrt{2}} = \frac{1}{\sqrt{2}} = 0.7071$$

4) $\boldsymbol{x} = (-\mathbf{2}, -\mathbf{1})$

$$z_1 = \frac{-3}{\sqrt{2}} = -2.1213, z_2 = \frac{(-2)-(-1)}{\sqrt{2}} = \frac{-1}{\sqrt{2}} = -0.7071$$

So, the score matrix $Z = [z_1 \ z_2]$is:

$$\boldsymbol{Z} = \begin{bmatrix} \mathbf{2.1213} & \mathbf{0.7071} \\ \mathbf{2.1213} & -\mathbf{0.7071} \\ -\mathbf{2.1213} & \mathbf{0.7071} \\ -\mathbf{2.1213} & -\mathbf{0.7071} \end{bmatrix}$$

Step 6) (Optional) 1D compression using only PC1 + reconstruction

If we keep only PC1, we approximate:

$$\hat{\boldsymbol{x}}_{\boldsymbol{i}} = \boldsymbol{z}_{\mathbf{1},\boldsymbol{i}} \boldsymbol{v}_{\mathbf{1}} (\textbf{since } \boldsymbol{\mu} = \mathbf{0})$$

Example for $x = (2,1)$, $z_1 = \frac{3}{\sqrt{2}}$:

$$\hat{x} = \frac{3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix}$$

Reconstruction error:

$$e = x - \hat{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$$

This shows: **PC1 captures the "main diagonal trend"**, while **PC2 captures the small deviation** between $x_1$and x_2.

Question 3:

Life example: combining Math & Physics into one "STEM ability" score

Suppose you have **4 students** with scores (out of 20) in **Math** and **Physics**:

$$\boldsymbol{X} = \begin{bmatrix} \mathbf{12} & \mathbf{11} \\ \mathbf{11} & \mathbf{12} \\ \mathbf{9} & \mathbf{8} \\ \mathbf{8} & \mathbf{9} \end{bmatrix}$$

Rows = students ($S1$, $S2$, $S3$, $S4$), columns = (Math, Physics).

Goal: replace 2 correlated marks with:

- **PC1:** overall STEM performance (one number)
- **PC2:** "Math vs Physics tilt" (who is stronger in which)

Step 1) Compute the mean of each subject

Math means:

$$\mu_1 = \frac{12 + 11 + 9 + 8}{4} = \frac{40}{4} = 10$$

Physics mean:

$$\mu_2 = \frac{11 + 12 + 8 + 9}{4} = \frac{40}{4} = 10$$

So:

$$\mu = [10,\ 10]$$

Step 2) Center the data

$$X_c = X - \mu$$

Student-wise:

- $S1: (12,11) - (10,10) = (2,1)$
- $S2: (11,12) - (10,10) = (1,2)$
- $S3: (9,8) - (10,10) = (-1,-2)$
- $S4: (8,9) - (10,10) = (-2,-1)$

$$X_c = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{bmatrix}$$

Step 3) Covariance matrix

For $n = 4$,

$$\boldsymbol{C = \frac{1}{n-1} X_c^T X_c = \frac{1}{3} X_c^T X_c}$$

Compute the needed sums:

Variances

$$\sum x_1^2 = 2^2 + 1^2 + (-1)^2 + (-2)^2 = 4 + 1 + 1 + 4 = 10$$
$$\sum x_2^2 = 1^2 + 2^2 + (-2)^2 + (-1)^2 = 1 + 4 + 4 + 1 = 10$$

Covariance term

$$\sum x_1 x_2 = (2)(1) + (1)(2) + (-1)(-2) + (-2)(-1) = 2 + 2 + 2 + 2 = 8$$

So:

$$C = \frac{1}{3}\begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} & \frac{8}{3} \\ \frac{8}{3} & \frac{10}{3} \end{bmatrix}$$

Step 4) Eigenvalues (variance along PCs)

This matrix has the form $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$, so eigenvalues are:

$$\lambda_1 = a + b, \lambda_2 = a - b$$

Here $a = \frac{10}{3}, b = \frac{8}{3}$:

$$\lambda_1 = \frac{10}{3} + \frac{8}{3} = \frac{18}{3} = 6$$
$$\lambda_2 = \frac{10}{3} - \frac{8}{3} = \frac{2}{3}$$

Explained variance

Total variance:

$$\lambda_1 + \lambda_2 = 6 + \frac{2}{3} = \frac{20}{3}$$

$$\mathrm{EVR}_1 = \frac{6}{20/3} = \frac{18}{20} = 0.90 = 90\%$$
$$\mathrm{EVR}_2 = \frac{2/3}{20/3} = \frac{2}{20} = 0.10 = 10\%$$

PC1 explains 90% of the variation → one score (PC1) summarizes performance very well.

Step 5) Eigenvectors (directions)

For $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$:

- eigenvector for $\lambda_1 = a + b$is $[1, 1]^T$
- eigenvector for $\lambda_2 = a - b$is $[1, -1]^T$

Normalize both:

$$v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Interpretation:

- **PC1 direction** $\propto (1,1)$: "Math + Physics together" → overall STEM ability.
- **PC2 direction** $\propto (1, -1)$: difference "Math − Physics" → subject preference.

Step 6) Compute PCA scores (project each student)

Scores are:

$$\boldsymbol{z_{1,i} = x_{c,i} \cdot v_1, z_{2,i} = x_{c,i} \cdot v_2}$$

Since $v_1 = \frac{1}{\sqrt{2}}[1,1]$:

$$z_1 = \frac{x_1 + x_2}{\sqrt{2}}$$

and $v_2 = \frac{1}{\sqrt{2}}[1, -1]$:

$$z_2 = \frac{x_1 - x_2}{\sqrt{2}}$$

Now calculate for each student (using centered pairs):

Student 1: $(2, 1)$

$$z_1 = \frac{2+1}{\sqrt{2}} = \frac{3}{\sqrt{2}} = 2.121 z_2 = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}} = 0.707$$

Student 2: $(1، 2)$

$$z_1 = \frac{3}{\sqrt{2}} = 2.121 z_2 = \frac{-1}{\sqrt{2}} = -0.707$$

Student 3: $(-1, -2)$

$$z_1 = \frac{-3}{\sqrt{2}} = -2.121 z_2 = \frac{(-1)-(-2)}{\sqrt{2}} = \frac{1}{\sqrt{2}} = 0.707$$

Student 4: $(-2, -1)$

$$z_1 = \frac{-3}{\sqrt{2}} = -2.121 z_2 = \frac{-1}{\sqrt{2}} = -0.707$$

So the score matrix is:

$$\boldsymbol{Z} = \begin{bmatrix} \mathbf{2.121} & \mathbf{0.707} \\ \mathbf{2.121} & \mathbf{-0.707} \\ \mathbf{-2.121} & \mathbf{0.707} \\ \mathbf{-2.121} & \mathbf{-0.707} \end{bmatrix}$$

Final teaching takeaway (very clear)

- **PC1 (90%)** is basically: $z_1 \propto (\text{Math} + \text{Physics})$→ use it as a **single "overall STEM score."**
- **PC2 (10%)** is basically: $z_2 \propto (\text{Math} - \text{Physics})$→ tells you who is **more Math-leaning vs Physics-leaning**.

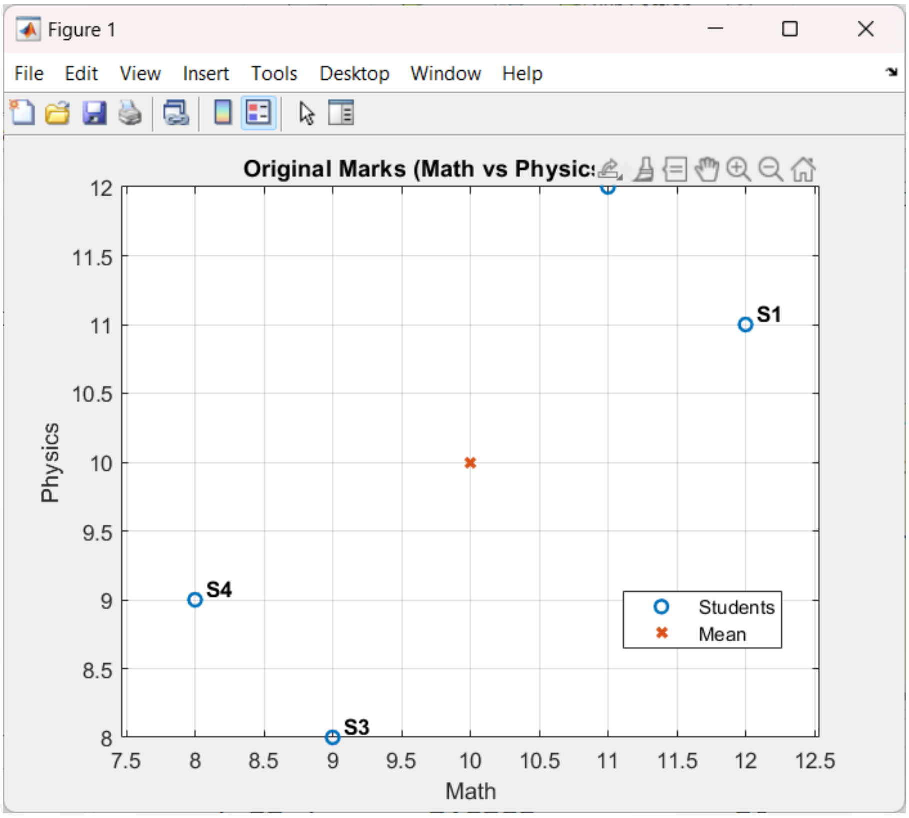
Figure 1
File Edit View Insert Tools Desktop Window Help
Original Marks (Math vs Physic
Physics
Math
S1
S3
S4
Students
Mean
12
11.5
11
10.5
10
9.5
9
8.5
8
7.5
12.5

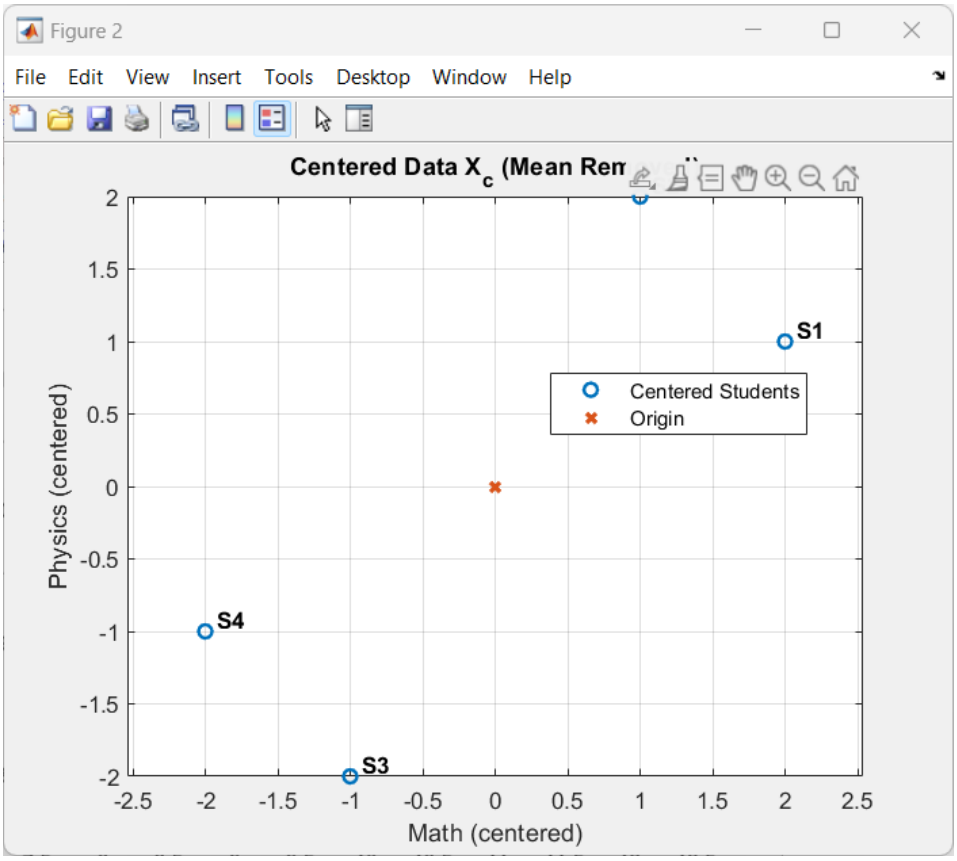
Figure 2
File Edit View Insert Tools Desktop Window Help
Centered Data X c (Mean Rem
Physics (centered)
Math (centered)
S1
S3
S4
Centered Students
Origin
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
2.5

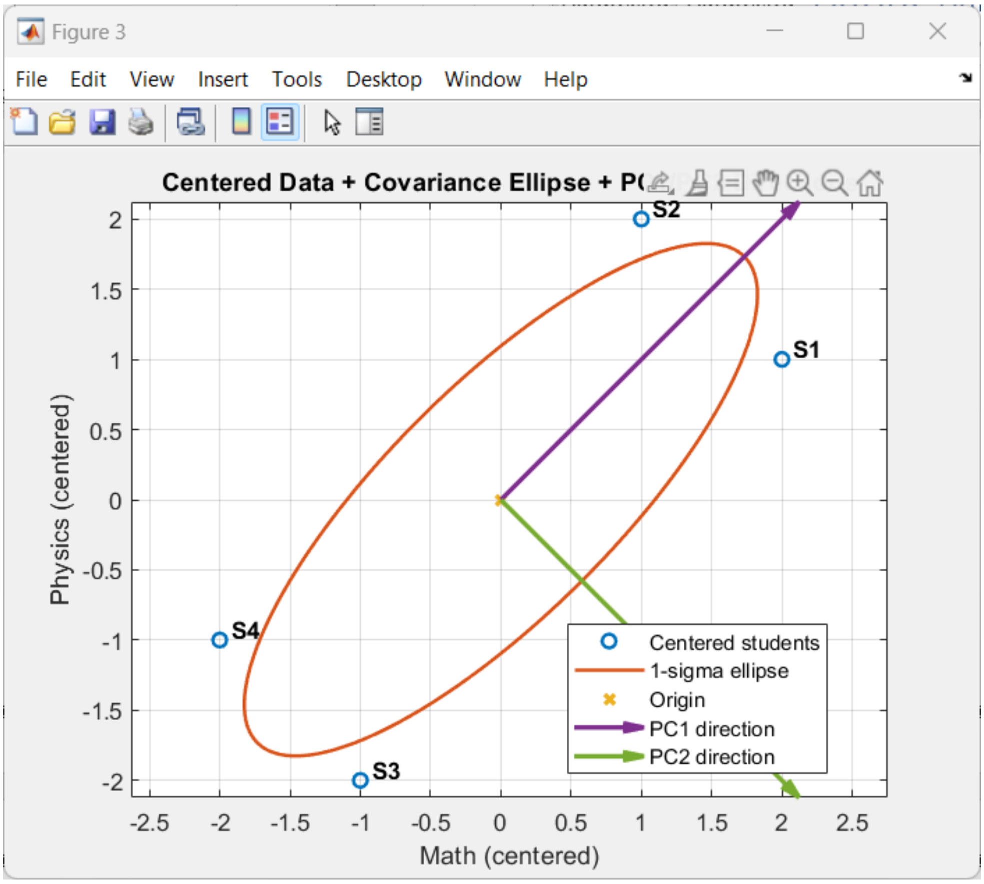
Figure 3
File Edit View Insert Tools Desktop Window Help
Centered Data + Covariance Ellipse + P(
S1
S2
S3
S4
Physics (centered)
Math (centered)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
2.5
Centered students
1-sigma ellipse
Origin
PC1 direction
PC2 direction

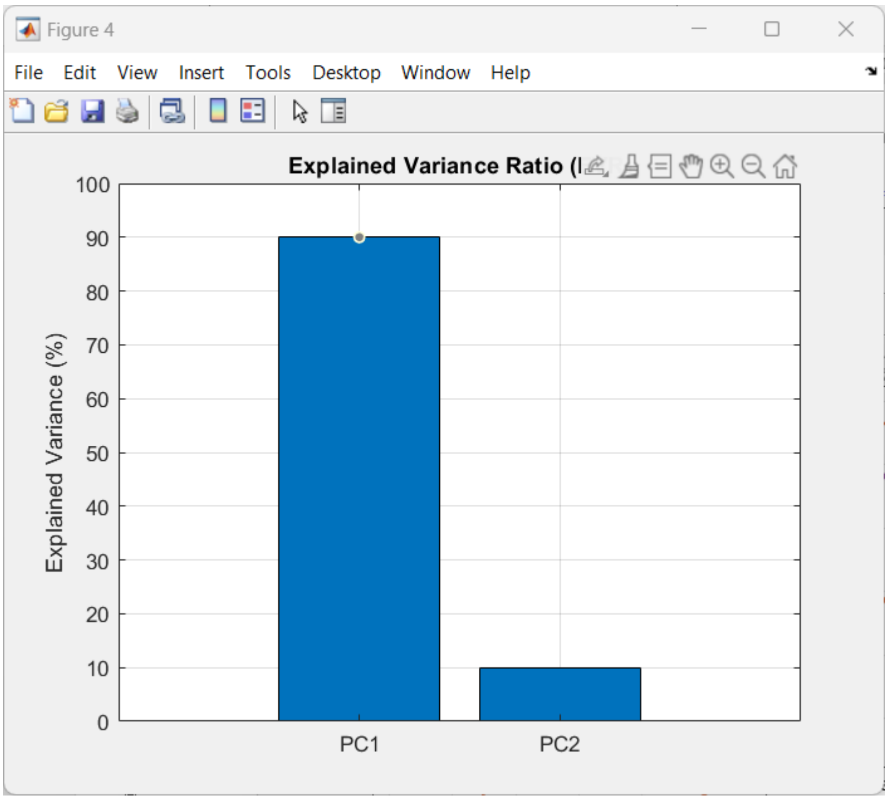
Figure 4
File Edit View Insert Tools Desktop Window Help
Explained Variance Ratio (
Explained Variance (%)
100
90
80
70
60
50
40
30
20
10
0
PC1
PC2

Sheet 6 Solution - (PCA)

==